

Automating Life Cycle Assessments through Artificial Intelligence Agents and Integrated Assessment Models

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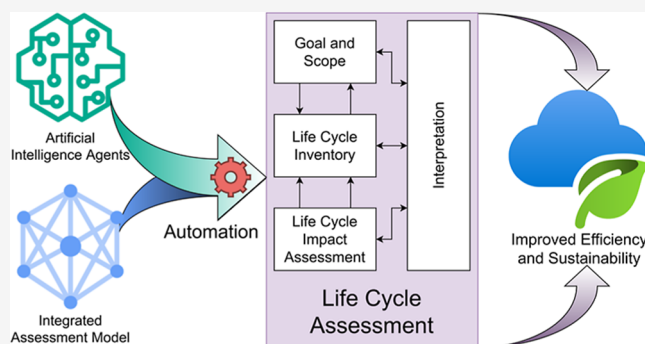


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ABSTRACT: Life cycle assessments (LCA) are a critical decision support tool for environmentally sustainable decision-making, but barriers such as time and resource intensity inhibit widespread application of LCA. To overcome these challenges, LCAs have been partially automated with integrated assessment models (IAM) and artificial intelligence (AI) agents, but existing literature reviews do not describe the potential for automation. We find that AI agents and IAMs offer tantalizing methods to automate all four stages of LCA with many avenues for future research, but adoption in the LCA community outside of the premise tool is minimal, despite the reliable results reported by the authors of several studies. Automated LCA can improve time and resource efficiency across all four stages of LCA, but there is insufficient research to rigorously assess the quality of automated LCA. To overcome the



KEYWORDS: life cycle assessment, large language models, integrated assessment models, artificial intelligence agents, environmental sustainability

1. INTRODUCTION

The world is in a sustainability crisis; between climate change,¹ widespread eutrophication,² and biodiversity loss,³ among others, there is a need to make sustainable choices and develop and use sustainable products to avoid environmental harm. The most common method to assess the environmental sustainability of products and services is using a tool called life cycle assessment (LCA), which is standardized in ISO 14040 and 14044.⁴ LCAs can be integrated into decision-making processes for governments, corporations, and everyday people,^{5,6} which can further enable the development of an environmentally sustainable world. While LCAs are powerful tools, they suffer from high time and resource intensity, limiting their widespread applicability as decision support tools to help decision-makers make more environmentally sustainable decisions.

In this age of available and cheap computing power, integrated assessment models (IAM) and artificial intelligence (AI) agents are powerful tools that can facilitate the development of automated LCA frameworks. IAMs provide a description of key processes in earth and human systems, typically focusing on climate, energy, water, land, and socioeconomic systems and projecting these processes forward in time for harmonious future predictions.⁷ Associated IAM tools like premise⁸ leverage these future scenarios to automatically project life cycle inventory (LCI) data forward

in time, showing how the environmental impacts of products could change in response to various policy scenarios. Moreover, there is potential to fully integrate LCA into an IAM, automatically completing LCAs for all products and processes available in an IAM, potentially vastly reducing the time to conduct LCAs when compared to a manual approach leveraging premise. AI agents can be defined as “artificial entities that sense their environment, make decisions, and take actions”.⁹ AI agents have been successfully applied to a variety of domains, including sustainability studies;¹⁰ for example, AI agents are currently used to automatically map bills of materials to LCI database entries and provide basic LCA support, enabling LCA practitioners to partially automate LCA. By utilizing these AI agents across all LCA tasks, LCAs could be reliably completed at scale with a significant reduction in time, resources, and cost. An LCA framework automated with AI agents would be able to complete an LCA in minutes, not months, that many such projects typically take when completed manually. Given these opportunities, it is critical to

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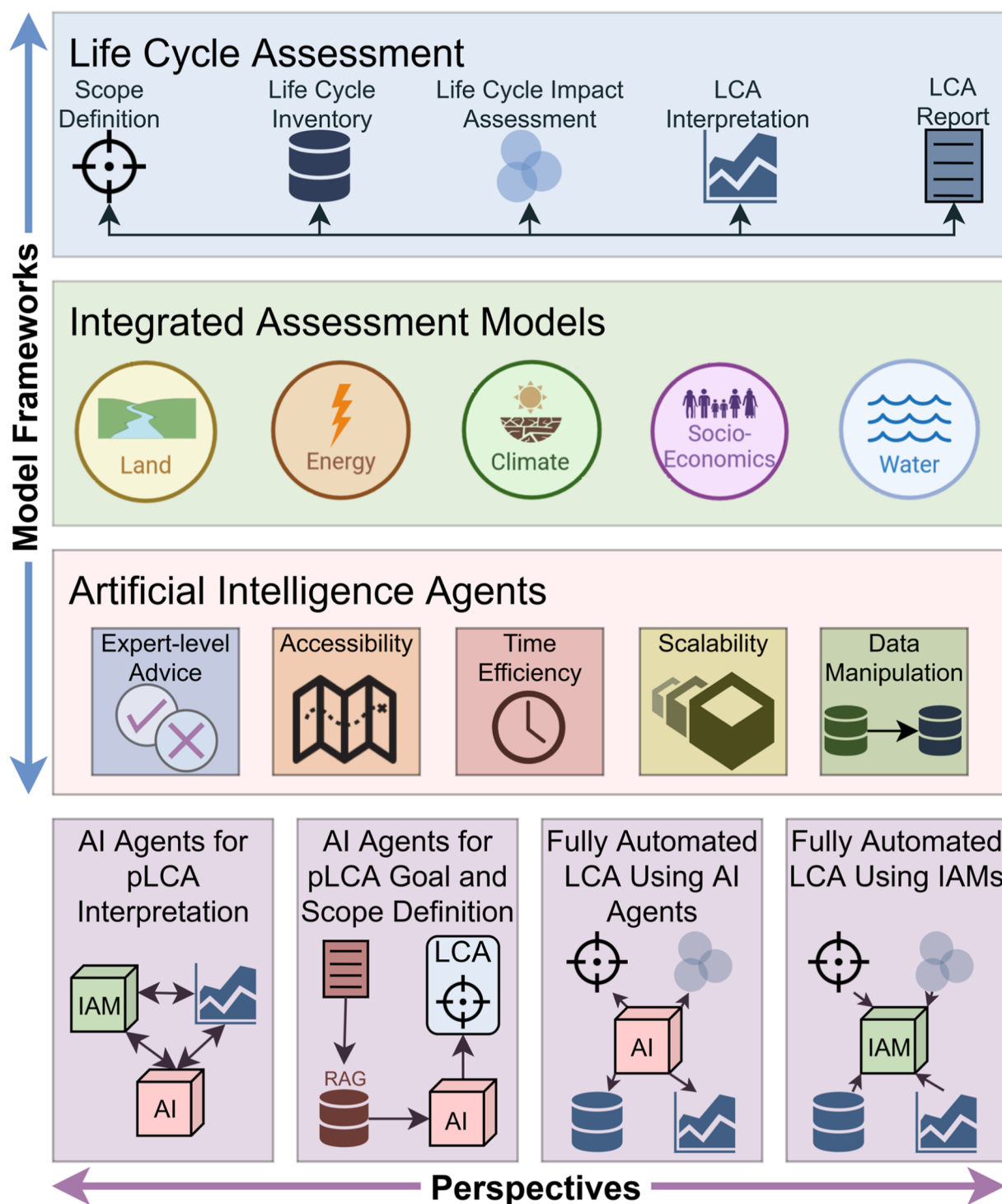


Figure 1. Overview of the computational frameworks contained in the review and future perspectives for the adoption of these frameworks. Abbreviations: IAM—Integrated Assessment Model, LCA—Life Cycle Assessment, RAG—Retrieval Augmented Generation, AI—Artificial Intelligence, and pLCA—prospective Life Cycle Assessment.

identify other advantages and limitations resulting from automated LCA frameworks. When combined into a single framework, IAMs, AI agents, and LCA hold additional promise over frameworks that include only two of the three: additional time savings when linking LCA and IAM with AI agents and a more robust evaluation of the future performance of technology when using IAM scenarios in LCA and AI agent studies.

A systematic evaluation of automated LCA has not been conducted. Comparison of automated LCA and IAM frameworks, or prospective LCA (pLCA), in the context of automation with AI agents is not present in the existing literature. Of the related pLCA reviews, one focuses on providing specific methodological guidance on pLCA but does not focus on automation and only contains minimal discussion of AI methods.¹¹ When reviews focus on linking IAM and LCA more broadly, the focus of analysis remains on scenario

generation¹² or methodological improvements for IAMs¹³ instead of automation, a gap that is addressed in this study. There is no specific review on AI agents and LCA, but there is related work on large language models (LLM), a model commonly used in agentic AI and LCA frameworks. These studies give an overview of the potential of LLMs to improve LCA methods but does not include a comprehensive review of existing technical studies and includes only a minimal discussion of automation.¹⁴ Another study, focusing solely on LLMs for LCI, does not contain a comprehensive literature review and also contains minimal discussion on automation.¹⁵ Moreover, none of these studies explore the nexus of AI agents and IAMs for the automation of LCA, which is addressed in this paper. There is a knowledge gap in highlighting the strengths, limitations, and capabilities of state-of-the-art automated LCA frameworks. Even partially automating LCA could provide significant benefits for using LCAs as the basis for environmentally sustainable decision-making, but these frameworks may require the development of norms in the LCA community to ensure that they are used in a transparent and trusted manner.

2. BACKGROUND

LCA is the state-of-the-art method for analyzing the environmental impacts of a product or service. In Section 2.1, a brief overview of the four stages of LCA is given, followed by a discussion of related LCA tools. This is meant to be a high-level summary; readers looking for an in-depth guide to LCA methods are encouraged to consult the following sources.^{16,17} Additionally, this section gives an overview of two of the largest and most powerful model classes that can augment LCA methodologies: IAMs and AI agents. These models are complex and large in computational terms—numbers of parameters, model run time, memory requirements, etc.—and offer significant benefits in improving the ability of LCAs to be automated. IAMs provide holistic and self-consistent scenarios for analysis, capturing the delicate links between the background and foreground systems in LCA at a high level.⁸ AI agents significantly decrease the time necessary to conduct LCAs,^{10,18} without necessarily sacrificing accuracy or reliability.^{19,20} An overview of these models can be found in Sections 2.2 and 2.3. An overview of these models can be found in Figure 1. In Section 2.4, definitions of automation are discussed.

2.1. Life Cycle Assessment. LCAs are powerful tools, standardized yet flexible. The ISO 14040 framework describes the four main stages of an LCA: goal and scope definition, LCI, life cycle impact assessment (LCIA), and interpretation.²¹ LCAs have been adapted to meet a wide range of modeling questions and will continue to evolve through the confluence of powerful computational models such as IAMs and AI agents. In this subsection, the four stages of an LCA are described, followed by a discussion of common LCA frameworks. Finally, some of the shortcomings of LCA are described and how IAMs and AI agents can automate LCA methods to overcome these challenges.

In the goal and scope definitions, the LCA practitioner defines the major parameters of the study. They define the goal or purpose of the study, giving additional information that determines how the study is conducted and how the results are communicated.²² The scope of the LCA determines the product systems under assessment and the methods by which the products are assessed.²³ The goal and scope inform the

type of LCA to be conducted, with consequential LCA preferred by some²⁴ and attributional LCA by others²⁵ when using LCA as a decision support tool. The goal and scope definition serve as a foundation for the later phases, but the decisions made in this phase can be retroactively updated as the LCA practitioner learns more about the system of interest. Setting a system boundary and completing goal and scope tasks require time and expertise—AI agents can substitute rapid evaluation and synthesis of existing studies for expertise, giving the LCA practitioner near-instant access to the best practices of LCA in this sector. This could result in increased standardization of goal and scope topics, as the best practices are automatically applied to future studies, facilitating comparisons of results between LCAs.

In the LCI, the LCA practitioner collects data and models flows for all processes and products which are inside the system boundary.²⁶ Oftentimes, this is an iterative process, as the information uncovered while finding the LCI is used to update the system boundary created in the goal and scope phase.²¹ These data are split into two categories: foreground and background. While foreground LCI data are often taken firsthand by the LCA practitioner, they can also be derived from process data, simulations, or literature data. Background LCI data is often found in databases like ecoinvent and describes inputs to the system of interest.²⁷ Because flows in the system of interest cannot always be directly matched to a flow in a database, it requires considerable time and energy for the LCA practitioner to collect the necessary LCI data. Partial automation by AI agents will save the LCA practitioner considerable time on these tasks, especially for background data that are commonly taken from databases. Meanwhile, data from IAM scenarios are already automatically incorporated into LCI databases using premise, facilitating the use of pLCA.

In the LCIA, calculations are conducted to calculate the environmental impacts of the product or process being evaluated by the LCA.²⁸ LCI data are matched with impact factors, which describe the size of the impact each flow has on the environment. While standalone indicators like global warming potential calculated using the IPCC 2013 method²⁹ do exist, it is common for many impact categories to be holistically combined in frameworks such as ReCiPe,³⁰ ILCD,¹⁷ and IMPACT 2002+³¹ to evaluate the product system under a holistic set of environmental sustainability indicators. Typically, open-source or commercial software such as OpenLCA³² is used to automate LCIA calculations. Once these impacts are calculated, they are used as the basis for the analysis in the interpretation phase of an LCA, which is also partially automated using existing software. When integrating LCA with IAMs, environmental impacts could be calculated as the IAM solves for future scenarios, showing great promise for automating the LCA.

In the interpretation phase of the LCA, the authors of the study interpreted the results of the study in the context of the uncertainties and assumptions used. These interpretations are drawn on the material gathered and the information learned in previous steps. Often, practitioners look for environmental hotspots in a product or process or else compare different products to choose the most environmentally sustainable one.³³ Then, recommendations and conclusions are drawn based on the results of the study and related to the audience of the study identified in the goal and scope definition. Based on the context of the study, the LCA may then be used as the basis for decision-making, enabling the selection of more

environmentally friendly products and services. This stage benefits from integrating AI agents by automating some of the common interpretation tasks such as hotspot identification, saving the LCA practitioner time and enabling them to focus on more complex analysis and synthesis of results. Moreover, by fully integrating LCA into the IAM, IAMs can solve for future sustainability scenarios that are not dependent upon avoiding climate change, enabling the potential to focus on critical areas such as eutrophication or ecotoxicity.

As more complex sustainability questions have been asked, the LCA framework has evolved and changed to better be able to answer these questions. The basic type of LCA is attributional, which is widely applied and established³⁴ and serves as a useful baseline against which to compare other LCA methods. In an attributional LCA, the environmental impacts of flows to and from the system of interest are described and calculated.³⁵ In comparison, in a consequential LCA the environmental impacts caused are as results of the decision to produce the product in the system of interest,³⁵ using marginal instead of average suppliers to assess the environmental impacts of products.³⁶ LCAs examining the future impacts of technologies, called pLCA or ex-ante LCA,^{37,38} are typically based on attributional LCA methodologies but estimate the impacts of the system of interest in the future based on various scenarios commonly generated by IAMs.³⁹ Furthermore, LCA has been modified to account for social instead of environmental sustainability, measuring the social impacts of the system of interest.⁴⁰ Tools based on LCA methods like product carbon footprints^{41,42} and environmental product declarations (EPD)⁴³ are also used as the basis for environmentally sustainable decision-making, and the findings in this study could similarly be applied.

2.2. Integrated Assessment Models. IAMs are complex earth and economic systems models primarily used to assess the impacts of climate change, identifying the basket of technologies and policies that lead to more sustainable and lower carbon futures.^{44,45} There are numerous IAMs, each considering different economic systems, temporal and geographical scopes, greenhouse gases, and environmental processes. Five common IAMs IMAGE,⁴⁶ MESSAGE-GLOBIOM,⁴⁷ AIM/GCE,⁴⁸ GCAM,⁴⁹ and REMIND-MAGPIE⁵⁰ are used as baseline scenarios for the shared socioeconomic pathways (SSP) and have publicly available output data,⁵¹ increasing the accessibility of these models for LCA practitioners. IAM scenarios improve LCA predictions by offering a harmonious, internally consistent set of assumptions and future economic projections based on those assumptions, which eliminates the need to combine projections from disparate sources. IAM scenarios reduce uncertainty in pLCA by modeling several scenarios under vastly different assumptions, offering guidance about the likelihood of the future state of the economy and environment. To facilitate comparisons between IAMs, model metadata exists,⁵² which can aid pLCA practitioners in selecting the appropriate IAM and IAM-generated scenarios. IAMs have been used to identify the decarbonization strategies necessary to restrict global warming to 1.5 °C by 2100⁵³ and the conservation strategies necessary to reverse biodiversity loss caused by habitat conversion,⁵⁴ showing their success in sustainability policy applications.

To develop holistic, varied, and internally consistent assumptions, integrated assessment modelers developed the SSPs, a set of 5 scenarios with internally consistent

assumptions about future socioeconomic development.⁵¹ These scenarios are commonly paired with representative concentration pathways (RCP), which describe future levels of climate forcing.⁵⁵ Taken together, the two sets of pathways form the basis for about 25 scenarios commonly used by IAMs, though non-climate-related policies can also be formulated and modeled in IAMs.⁵⁶ Some SSP and RCP pathway combinations are commonly used as future scenarios in pLCA via premise, although there is room for further expansion of scenarios, as discussed in Section 4.2.

2.3. Agentic AI. AI agents make decisions and take actions autonomously using data and an AI model as the basis for the decision-making process. Any type of AI model can be used in an AI agent, from general machine learning (ML) and AI tools to reinforcement learning (RL) agents, to LLMs and multimodal foundation models. General AI tools are based on ML algorithms like k-nearest neighbors, random forest, support vector machines, XGBoost, and others.⁵⁷ Because the uses of general AI tools have been recently discussed in reviews,^{58,59} they are mostly omitted here, except for a few locations where these tools demonstrate superior capabilities over emerging AI agents. RL agents learn behavior through trial-and-error interaction with the environment.⁶⁰ When the agent makes a good decision, it is rewarded, and when it makes a bad decision, it is punished. By iterating over time, the agent attempts to learn the optimal policy for decision-making.⁶¹ Very few LCA studies leverage RL agents; there is a vast unexplored space for the capabilities of these agents to complete LCA tasks.

A more common model class underlying AI agents is LLMs, sophisticated state-of-the-art tools to mimic language that have achieved human-expert level accuracy across a variety of natural language processing (NLP) tasks.^{62–64} NLP tasks include question answering, sentence completion, translation, embedding, and named entity recognition.⁶⁵ Most LCA tasks can be reformulated as NLP tasks requiring specialized domain knowledge, highlighting the transferability of the advances in the computer science LLM literature to LCA tasks. One of the first LLM for LCA studies used NLP mapping techniques with the BERT language model⁶⁶ to match processes to an LCI database in 2023,⁶⁷ methods which have been adopted and expanded in subsequent years. Like IAMs, LLMs are computationally complex, requiring significant computer hardware to run. LLMs are constructed primarily on transformers,⁶⁸ a model architecture that has since spawned many variants.⁶⁹ If a custom LLM has been constructed, the next step in model creation is training,⁷⁰ for off-the-shelf models fine-tuning⁷¹ or prompt engineering⁷² may be more appropriate to improve performance on LCA tasks. Training, fine-tuning, and prompt engineering often require examples of the task at hand, requiring data engineering and other data science techniques to create a relevant data set or benchmark. LLMs are often evaluated on the accuracy of the task, though there are a variety of metrics, including qualitative analysis by humans, several of which are specific to certain types of tasks.⁷³ For further information on LLMs, the interested reader is directed to these reviews.^{74,75} Multimodal foundation models are similar in scale and scope to LLMs, but can process inputs in a variety of modalities, including vision and audio in addition to text.⁷⁶ While we are aware of a study that uses multimodal AI agents for LCA tasks,⁷⁷ it has not been peer reviewed so it is excluded from our analysis. Further research

should be conducted on integrating additional data modalities into autonomous LCA systems.

2.4. Automation of Life Cycle Assessment. For the automation of LCA, there are two main types: partial and full. Because different LCAs will report information for different tasks, it may be difficult to label a system as full automation, as it might not complete some tasks that an LCA practitioner would consider essential to conducting an LCA. As a result, it could be easier to conceptualize partial automation as a system that can be used by an LCA practitioner to automatically complete an LCA task, while full automation relegates the LCA practitioner to a supervisory role or eliminates the need for a human to complete an LCA.

3. LITERATURE REVIEW METHODS

Relevant articles on automated LCA frameworks were retrieved by the Web of Science Core Collection, Scopus, and these 3 EbscoHost databases: Agricola, Environment Index, and Academic Search Premier because they index journals that focus on sustainability and LCA. To the best of our knowledge, most of the existing literature on AI agents and LCA or IAM and LCA frameworks does not mention automation by name, requiring a less specific search string. The search string used in this study is as follows:

TS = (“life cycle assessment” OR “lifecycle assessment” OR “life cycle analysis” OR “lifecycle analysis” OR “cradle-to-grave analysis”) AND (“large language models” OR “natural language processing” OR “AI agent” OR “agentic AI”) OR (“Integrated Assessment Model” or “PREMISE”))

The search retrieved 161 articles from Web of Science, 272 articles from Scopus, and 73 articles from EbscoHost, for a total of 506 papers. These articles were then filtered to ensure that the paper should focus on the use of agentic AI or IAM augmented LCA frameworks, with promise for automation. Papers that utilized the premise tool were checked for methodological innovations—if premise was just used to develop a LCI database, then the papers were excluded as beyond the scope of this analysis. Due to the fast-evolving nature of agentic AI, conference papers were included in the sample. Studies published in journals with rapid publication timelines that may compromise peer-review stringency were not considered. In addition, references were snowballed from seminal works and review papers.^{12,14} This resulted in 75 papers that were included in further analysis. Additional methods are presented in the Supplementary Methods section of the Supporting Information.

Qualitative analysis of the implementations of the frameworks and potential for automation is conducted, as shown in Section 4. Quantitative evaluation of the studies on the year of publication and the stages of an LCA influenced by AI agents can be found in the Supplementary Results section of the Supporting Information.

4. AUTOMATING LIFE CYCLE ASSESSMENTS

As discussed in Section 2, IAMs and AI agents are complex computational models gaining relevance in sustainability studies. With increasing interest in these models in recent years, analyzing how these models contribute to LCA automation is becoming increasingly relevant. While great strides in the LCA community have been made to adopt these frameworks, there is still much work to be done to standardize the methods used in automated LCA frameworks. To the best

of our knowledge, no study has yet integrated all three model classes; therefore, the analysis of the state-of-the-art methods for these frameworks is conducted separately, though initial steps for developing integrated frameworks are proposed in Section 4.3. The main opportunities and limitations of these frameworks for automation are summarized in Table 1. Section 4.1 reviews AI agents for LCA automation; Section 4.2 reviews the IAM tools for LCA automation; and Section 4.3 contains summary tables listing the major available tools.

Table 1. Opportunities and Limitations for Automated LCA Frameworks

	IAM and LCA	AI agent and LCA
opportunities	•full integration of LCA into IAM for automatic LCA •automatic mapping of IAM scenario outputs into LCI database modifications	•improve AI agents for each task across the four stages of LCA •compare results from AI agent-automated LCA with LCAs completed by humans
limitations	•simplified system boundary representations •double counting of emissions •data challenges •high computational resource requirements •uncertainty in assumptions	•potential for LCA methodological errors •trustworthiness •accessibility •high computational resource requirements

4.1. Automation of Life Cycle Assessment Using Agentic AI. AI agents typically automate tasks within a single LCA stage, although some frameworks fully automate the entire LCA process. It has been widely documented that LLMs and related NLP processes contain “reliable” results compared to manual LCA processes,^{18–20,67,78,79} though these frameworks have mostly been developed for the LCI and interpretation stages. Less common is the use of RL agents for automating LCA tasks, and using general AI tools for LCA automation is outside the scope of this review. In this section, the utilization of AI agents is analyzed for each stage of the LCA, followed by a discussion of the drawbacks of AI agents for automating the LCA.

4.1.1. AI Agent Model Selection. Model selection for AI agents is driven primarily by the availability. Most of the AI agent frameworks evaluate models developed by OpenAI, typically whichever version of GPT⁶³ was most recent at the time of publication,⁸⁰ in part due to the ease of access and general awareness of ChatGPT. Frameworks incorporating retrieval augmented generation (RAG) outperformed off-the-shelf GPT and Qianwen⁸¹ models,²⁰ while Gu et al. report that Claude 3.5⁸² outperformed GPT-4o on LCI matching tasks¹⁸ and Luo et al. report that GPT models outperformed Qianwen.¹⁹ In the most comprehensive comparative analysis of LLMs in an LCA framework, Balaji et al. report that Claude models outperform Llama 3.1 Instruct 70B⁶⁴ on recommending impact factors and also evaluate the performance of an array of embedding models.¹⁰ Moreover, Zhang et al. argue that a data abstraction layer enables custom models to outperform an off-the-shelf GPT-4.1 model on both robustness and reproducibility of environmental impact estimates.⁸³ As of now, it is difficult to determine which off-the-shelf LLM should be used for each task. It would be beneficial for the LCA community to evaluate which AI agents and which additional layers such as RAG, data abstraction, knowledge graphs, etc.,

enable the best performance on automating LCA tasks, necessitating the use of comparative studies.

4.1.2. AI Agents for the Four Life Cycle Assessment Phases. As mentioned in the supplementary results, the goal and scope definition stage of an LCA is understudied in the AI agent and LCA literature, in large part due to data challenges because of a lack of benchmarking data sets. AI agents show promise for devising system boundaries for simple LCA systems that contain only a few processes,⁸³ though more complex boundaries will require more experience and expert-labeled high-quality data. NLP methods have shown great promise at summarizing and synthesizing stakeholder perceptions across multiple settings including focus groups, brainstorming, semistructured dialogues, and formal surveys to identify impact categories of high relevance to LCA studies.⁸⁴ However, the authors caution that NLP techniques may only conduct surface-form word matching, losing deeper contextual meaning and causality—a data challenge. Difficulties with data requirements are exacerbated by the proliferation of LCA frameworks and the lack of consistent reporting of ISO-required information in the goal and scope definition phase in the existing body of LCA studies, complicating the use of nonexpert labelers to generate a benchmarking data set. Because there are few studies on AI agents for goal and scope definition tasks, it may appear as if there are many shortcomings before AI agents can automate this stage of the study. However, given the great corpus of existing LCA studies and related works, this area is ripe for further research on frameworks synthesizing NLP methods and qualitative stakeholder interactions with formalized data sets and general AI models for complex goal and scope tasks.

The benefits of using AI agents for LCI tasks are many, as LCA tasks such as extracting LCI data from other data sources, mapping flows to an LCI database, generating LCI data, or helping researchers understand LCI databases have all been partially automated. AI agents are used to automate mapping flows in a study^{10,14,15,18,19,85} or outputs from building information model systems^{67,86} to products or processes in LCI databases, with some agents returning “no match” if a given flow is missing in an LCI database.¹⁰ To aid in the semantic matching of these technologies and improve accuracy, some are arguing for the development of a shared vocabulary between flows and products in the LCA and in the LCI database.⁸⁶ In the absence of such a vocabulary, AI agents can still achieve high accuracy for environmentally extended input–output LCAs by embedding information using LLMs, but accuracy is lower for process-based LCA.¹⁰ These AI agents rely heavily on LCI databases, which may lead to inaccuracies, as high-quality data germane to the context of the study is not considered by the AI agent. While AI agents have success extracting LCI data from text documents like EPDs,^{78,87–90} shortcomings appear when using the AI agent for processing long technical datasheets, complicated processes, and niche products.⁸³ AI agents can also generate impact factors for flows or products,¹⁵ but the choice of AI model has repercussions. LLMs may have difficulties in providing useful answers to questions beyond their training scope,⁹¹ so it might be better to rely on simpler and more explainable general AI tools that are commonly used for impact factor matching.⁵⁸ As an example, artificial neural networks have been used to evaluate the LCI of novel chemicals in the absence of more reliable information.⁹² LLMs can also aid in the data quality assessment⁷⁹ or validation¹⁵ of LCI data.

Finally, other studies have looked at the potential of using transformers as the basis of AI agents⁹³ or developing an AI agent to help LCA practitioners navigate LCI databases.⁹⁴ Because AI agents have demonstrated capabilities for partially automating LCI tasks, the next steps are to evaluate the capabilities across diverse case studies, identify strengths and weaknesses, and iterate upon AI agent design. Finding out in which contexts these agents are inaccurate, inefficient, or unwieldy requires designing these agents for LCA practitioners, developing use cases, and scaling up the technologies.

Because most LCIA calculations are automatically handled by dedicated third-party software, there is not an imperative need to use AI agents to conduct LCIA, though such a study exists.²⁰ Other studies have commented upon the suitability of using LLMs as the basis of AI agents for this step, remarking on the weakness of LLMs because they often do not conduct mathematical computations upon the available data.⁹⁰ Going forward, agentic AI may be given access to LCIA software, automatically set up product systems, and run calculations inside existing LCA software, eliminating the need for LCA practitioners to complete these tasks. However, it is also possible that AI agents will leverage standard mathematical libraries to perform matrix operations to conduct the LCIA.

In the interpretation phase of LCA, AI agents have immense potential, but studies have mostly relied on qualitative analysis to assess the abilities of AI agents to automate results analysis. It is widely documented that basic interpretation support is available using off-the-shelf LLMs.^{14,78,95,96} More complex analysis is also possible using AI agents, for example, helping an LCA practitioner optimize a product system by identifying hotspots^{93,96} or suggesting alternative materials or flows.⁷⁸ RL-based AI agents can also leverage results from an LCA in an optimization problem to lower the intensity of carbon emissions.⁹⁷ AI agents can also leverage LCA reports to do additional tasks, such as assessing the environmental risk of choosing a particular supplier⁹⁸ or detect greenwashing.⁹⁹ Other possibilities include autonomously conducting a sensitivity analysis and visualizing results;⁸³ though there are some inherent risks of using LLMs to write policy implications or make decisions based on the results of LCA.¹⁴ Alcaide-Marzal and Diego-Mas find that some LLMs may present generic recommendations making limited use of data specific to a study, but this issue can be ameliorated by using prompt engineering.¹⁰⁰ LCA practitioners may rely upon contextual knowledge and implicit criteria in the system of interest, which can cause misalignment between the interpretation of the AI agent and the human practitioner.⁸⁸ Additionally, Zhang et al. report that participants who use an automated LCA tool are concerned with data reliability and greenwashing in the interpretation of LCA results.⁸³ More study is needed to fully understand the capabilities of AI agents to produce high-quality interpretation of LCA results as well as how, when, and why these interpretations differ from those of LCA practitioners. More case studies with both human and AI agent interpretation of LCA results need to be conducted as well as meta-analyses comparing the disparate findings across the case studies. This will enable an evidence-based approach toward introducing AI agents in decision-making processes for environmentally sustainable systems.

4.1.3. Limitations on AI Agents for Life Cycle Assessment. There are some potential drawbacks to using AI agents to automate LCA, specifically environmental sustainability, lack of available data sets, bias and accountability, hallucinations and

explainability, and AI agent evaluation. Many of these potential flaws are complex and systemic; fixing them will require collaboration with the computer scientists that develop the models that underpin AI agents and the development of community norms regarding the reporting and use of AI agents in LCA methodologies.

AI agents, especially those that rely on LLMs, may not be environmentally sustainable. Because it is known that LLMs consume high quantities of energy,¹⁰¹ reporting the carbon intensity of an automated LCA framework is beneficial,⁸³ but an LCA needs to be conducted to identify the environmental impacts of LCA processes conducted by humans. As the AI models underpinning AI agents become smaller and resemble general AI tools, environmental sustainability improves as fewer computations are required to complete a task. LCA practitioners will need to evaluate the trade-offs between accuracy, efficiency, and environmental impacts of different AI agents to ensure that the means of attaining a more environmentally sustainable world are not in and of themselves unsustainable.

Apart from LCI databases, other LCA stages lack high-quality, expert-labeled data sets that are easily accessible by AI agents, leading to many calls for the development of LCA-specific data sets.^{14,15,19,79,84} The construction of these data sets will further enable automation of LCA by increasing the quality of responses from AI agents and reducing the rate of errors. Leveraging existing initiatives for harmonized LCA data sets like Hestia for agriculture¹⁰² or a recently developed data set for buildings¹⁰³ would help reduce the proliferation of competing standards. Moreover, leveraging existing LCA ontologies or graphs may improve the quality of response from AI agents and prevent the proliferation of competing frameworks. Using graphs as representation for LCI has been around since at least 2016,¹⁰⁴ and there are several proposals for LCA ontologies.^{105–108} Living Sustainability offers an example of an agentic AI tool that includes an ontology for LCA as a data abstraction layer to improve model performance.⁸³ Because not all domains and LCA tasks have these data sets, further collaboration building on the successes of existing data harmonization efforts in LCA and industrial ecology would be beneficial for the development of automated LCA using AI agents. However, it is important to consider regulatory considerations and data privacy—LCA practitioners should be careful and follow appropriate laws and regulations when constructing data sets and benchmarks to evaluate AI agents.

AI agents are not neutral. There is substantial computer science literature detailing the biases of the underlying models for AI agents,^{109–111} yet their use in LCA can be used to detect biases in LCA practitioners.^{79,84,88} Detection of bias can run both ways—care should be given to ensure that AI agents do not blindly confirm the viewpoint of a biased LCA practitioner but neither should humans automatically approve results from AI agents without checking for bias. The key is to understand in what contexts bias appears and the development of specific strategies to curtail the impact of this bias. As an example, the AI agent may act inadvertently as a lobbyist for certain stakeholder groups or worldviews that may not be relevant to or actively harmful to the context of the LCA study. As a result, it is recommended that LCA processes automated via AI agents have human supervision to ensure methodological rigor and check for potential biases in the interpretation of LCA results or human-in-the-loop.^{14,112} However, the use of

humans overseeing automated LCA could be an instance of an “accountability sink”.¹¹³ In a hypothetical scenario with a biased AI agent, the human checking the agent’s work could be held accountable for not catching the displays of bias, but the AI agent would not be held accountable and would require no changes to its decision-making process. While there is no perfect solution to this problem, ensuring that the developed AI agents are accessible to a broad audience and responsive to community feedback may ameliorate the issue of bias and accountability for AI agents.

Of the risks of using AI agents to automate LCA tasks, accuracy and explainability are most likely to impact methodological rigor. One risk of using LLMs as the basis of AI agents is that they can suffer from hallucinations, where models produce plausible-sounding yet incorrect information,^{114,115} potentially leading LCA practitioners astray. To address this issue, LLMs are commonly modified through techniques such as retrieval augmented generation (RAG),¹¹⁶ wherein an external, nonparameterized knowledge base containing domain-specific language is leveraged by the LLM to improve response quality and accuracy.¹¹⁷ One study finds that for custom AI agent frameworks, the issue is not that the utilized LLM is inaccurate or hallucinates but rather that the difference in performance is derived from how the LCA tasks are formulated and what the prompts instruct the AI agent to do (or not do).⁸⁸ It is possible that proper utilization of specialized domain knowledge will be as pertinent as improving the accuracy and explainability of these agents. Another potential solution is to improve the interpretability of LLM models,^{18,89,118} though this is likely in the domain of computer scientists and not LCA practitioners. Alternatively, error detection systems could be created to check the work of AI agents.⁹⁰ Of these methods, RAG is best suited for LCA tasks due to the minimal data requirements and computational resources. For general AI models, explainable AI tools¹¹⁹ such as SHAP¹²⁰ may prove beneficial to reduce the occurrence of errors.

Researchers have not coalesced around the best way to evaluate AI agents on LCI tasks, with some using metrics like accuracy, mean average percent error, and precision,^{10,18} while Tu et al. argue for considerations evaluating LLMs on relevancy, accuracy, completeness, and specificity for impact factor matching tasks.¹⁵ Qualitative analysis of AI agent performance by LCA experts is also common.^{14,84,88} Because AI agents can complete a variety of LCA tasks, with each task having different requirements for accuracy and quality, the development of a universal set of metrics will be difficult. Moreover, due to the epistemic uncertainty inherent in LCA, it is difficult to establish the ground truth for many LCA tasks, severely compromising the ability of benchmarks to accurately judge AI agent performance. If there is no ground truth, AI agents should be judged based on the decision they made and the accompanying justification, with an eye toward evaluating if it would be reasonable for a human LCA practitioner to propose this same justification. Computationally, this is not an easy task, but it might be worth considering leveraging learnings from model alignment to improve AI agent performance,¹²¹ perhaps leveraging reinforcement learning.¹²² The LCA community should coalesce around responsible standards for evaluation of the performance of AI agents, though challenges may remain, as not all LCA tasks are clearly defined for automation.

While AI agent usage in LCA is currently minimal, the LCA community would benefit from widespread reporting and justification of usage of AI agents, as well as the quality and reliability of the underlying LLM models, to increase rigor and identify the strengths and weaknesses of AI agents. At first, AI agents may not be accessible to all practitioners or all contexts, as LCA studies have reported difficulties with using LLMs that cannot understand the local language.⁸⁴ Eventually, the LCA community will need to develop clear standards for the usage and reporting of AI agents in LCA, especially as automated LCA frameworks are more frequently adopted. As an example, automated pipelines should ensure that pertinent information, such as scenario IDs, regions, year, method/version, and source artifact for every generated record, should persist through all computations and decisions to ensure auditability and reproducibility. This problem is exacerbated by the limited number of case studies currently applying AI agents for LCA automation, requiring the need to test agents in diverse industry contexts.⁸⁸ Collaboration, norm development, and methodological rigor are important concepts to drive the development of automated LCA by using AI agents.

4.2. Automation of Life Cycle Assessment Using Integrated Assessment Models. IAMs have traditionally been conceived as a tool to expand the scope of LCA, typically through their use in the development of future scenarios for pLCA. In this subsection, we discuss how the state-of-the-art framework premise contributes toward the partial automation of pLCA using IAMs by increasing the accessibility of prospective LCI (pLCI) databases. Full automation of LCA requires a rethinking of how IAMs are used for LCA, but several studies have begun integrating LCA into IAMs, enabling the complete calculation of LCA results for technologies contained within the IAM. This is followed by a discussion of the opportunities and limitations of these approaches and how other IAM and LCA frameworks have the potential for partial automation of LCA and the potential for a significant increase in the number of scenarios available to pLCA practitioners.

4.2.1. Premise. Premise provides coherent scenarios from IAMs as an input to LCA through automatic updates to LCI databases, improving the quality, transparency, and consistency of pLCA.⁸ The usefulness of this tool is reflected in recommendations for its use,¹²³ and another review study has documented that many recent pLCA studies use premise.¹¹ Originally designed for the REMIND⁵⁰ and IMAGE⁴⁶ IAMs, premise has expanded to include the TIAM-UCL^{124,125} IAM, and further integration with IAMs and scenarios is possible. Premise solves the time-related representativeness problem of LCI data,¹²⁶ as IAMs can be used to project LCI data forward in time to account for technological innovation,⁸ account for the turnover of capital stocks,¹²⁷ and address the temporal mismatch between production and recycling.¹²⁸ Premise offers pLCI databases for both attributional (cutoff)⁸ and consequential¹²⁹ LCA. This builds on earlier work linking IAM scenarios to LCA, making use of the wurst package to link the IAM¹³⁰ and the superstructure package to manipulate scenario files.¹³¹ Premise, therefore, is considered to be a tool to enhance the scope of LCA via the creation of pLCI databases from consistent scenarios. While this in and of itself is not automation, when an LCA practitioner uses premise and does not develop their own pLCI database, this could be considered the partial automation for pLCA.

Despite the great success of premise and related tools in partially automating the selection of IAMs and scenarios for pLCA studies, there is more work to be done to improve the quality of pLCI databases and the IAMs to which they are paired so that high-quality pLCI databases are automatically available to LCA practitioners. It is recommended to expand regional coverage,^{126,132,133} technological coverage,^{11,132,134} environmental coverage,¹³² and sectoral coverage^{11,132} of pLCI databases. Even if coverage of these topics in IAMs is expanded, IAMs will still need to be mapped to LCI databases (and vice versa), which can introduce more uncertainty due to different geographical resolution,¹³⁵ sectoral/technological differences,^{8,12,134,136} double counting of emissions,¹² data disaggregation,^{126,137} and additional uncertainties and challenges.¹³⁸ A potential solution to these pLCI data challenges is to utilize the pathways framework to avoid double counting of emissions.¹³⁹ Supposing that the data between IAMs and pLCI databases is harmonized, there still may be other limitations of pLCI databases. For example, pLCI databases in certain scenarios may not anticipate disruptive shifts in technology,⁸ could use out-of-date IAM data,¹³² or have mismatched emissions factors with the IAM from which data is extracted.¹³⁶ These challenges are not specific to the process of automating LCA but may need to be solved to ensure that when automated LCA frameworks in IAM are widespread, they are high-quality and maintain methodological rigor.

4.2.2. Full Automation of Life Cycle Assessment. Full automation of LCA would entail creating an IAM model that automatically conducts LCA on all products and processes contained in the IAM. It has been argued that IAMs should be extended to include more LCA data,^{13,126,127,132,134,137,140} and there are several nascent implementations of such a harmonized framework. Studies have calculated LCA coefficients for inclusion into IAMs using THEMIS¹⁴¹ as an LCA framework,¹³⁷ which has been integrated with multiple IAMs to predict the environmental impacts of energy system decarbonization.¹⁴² The TIAM-FR IAM is extended with LCI data in the electricity sector to automatically calculate life cycle material use.¹⁴³ The IMED IAM¹⁴⁴ has been modified to include estimations of air pollution and water and land use under an LCA methodology in a study of the cement industry.¹⁴⁵ While not strictly utilizing an IAM, the ENBIOS project links LCA into a model of multiscale integrated analysis of societal and ecosystem metabolism with a focus on the energy system.¹⁴⁶ Finally, another study improves integration of background and foreground data, such that changes in foreground systems modify the LCI databases acquired via premise.¹³⁶ These studies undertake a different approach to combining IAMs and LCA than that embodied in premise and as such are a promising research direction for future LCA automation studies. Most of these studies have limitations for fully automating LCA; they conduct LCA only for some of the activities represented in IAMs and a subset of available environmental impact categories, highlighting that there is significant future potential for more closely linking LCA to IAMs. A fully automated IAM and LCA framework would require significant upfront investment of resources to identify all system boundaries and address allocation issues but could be quite powerful and flexible, as an LCA for each product could be automatically generated for each IAM scenario. However, the payoff of being able to run LCAs for thousands of common products in several geographical regions automatically would be quite beneficial to the field of pLCA, let alone

the potential to change the objective functions for IAMs to optimize for environmental sustainability indicators that are not affected by climate change. A potential implementation of such a framework is discussed further in the supplementary discussion.

The fully automated LCA approach using IAMs encounters difficulties due to the technological limitations of IAMs: the granularity of the modeled system and the associated data challenges. A fundamental limitation of IAMs is that they are unable to accurately describe all technologies in the technosphere—a problem that needs to be addressed for the successful automation of LCA using IAMs. While substantial progress has been made in improving the granularity of IAMs to date, there are several salient modifications necessary to IAMs, especially to increase resolution.^{8,11,13,127,132,134} Several studies have argued that the representation of material transformation from raw material extraction^{11,147} to waste management, recycling, circularity, and the end of life phase^{11,127,135,148–150} needs to be improved. These improvements could not only reduce uncertainty in pLCA but also require significant work from IAM developers and may only lead to false levels of precision,¹⁵¹ compounding extant data issues. Because there is a need to update the data used to parametrize IAMs and LCA regularly, there is a significant potential to conduct LCA studies using out-of-date data, potentially leading to inaccuracies in analysis and flawed policy decisions. Because of these data challenges, for many activities poorly modeled in IAMs, there may be a need to rely upon commercial databases to facilitate the completion of an LCA. Additionally, it may prove beneficial to develop regional-focused IAMs to provide this level of granularity¹⁵² or represent material stocks and flows in IAMs.¹⁵² The resolution of these limitations will require significant collaboration with the IAM community to ensure sufficient integration of the LCA methodology in IAM models for full automation. It may be more feasible to gradually expand the capacity of IAMs, for instance, by starting with improved material representation, so that practitioners can learn by doing and quickly iterate on IAM frameworks that fully automate LCA.

4.2.3. Partial Automation of Life Cycle Assessment. Finally, there are a few other alternatives for partial automation of LCA using IAMs, including bidirectional calibration, implementing more damage functions in IAMs, optimization, and the introduction of other industrial ecology or socio-economic metabolism models. Bidirectional calibration of IAMs and LCAs ensures that emission factors are aligned to make both model classes more policy relevant.¹⁵³ Because LCIA provides more specific and varied impact pathways in damage functions than most IAMs, modeling impacts of non-climate change assessment categories are tricky; this can be mitigated by some extent through adding dose–response functions to a dynamic optimization IAM.¹⁴⁰ Another approach is to monetize environmental impacts in the IAM,⁸ enabling the IAM to internalize the environmental externalities in its representation of markets. Moreover, some IAM and LCA frameworks enable optimization of the system, selecting the optimal combination of net-zero technologies to achieve decarbonization goals.¹⁵⁴ Another potential fix is to turn LCA into a type of bottom-up IAM, with tighter integration of economic and ecosystem modeling in the LCA framework itself,¹⁵³ the reverse of integrating LCA into IAM for full automation. Alternatively, a new type of model could be developed to serve as a standard model of the biophysical basis

of society,¹³ potentially including other socio-economic metabolism models beyond IAM and LCA.¹² De Bortoli et al. recommend coupling environmentally extended input–output models with IAMs.¹⁵⁵ These various approaches offer a wide range of potential research directions but are less promising for full automation of LCA due to decreased coupling between IAM and LCA, as well as further divergence from ISO-specified LCA methods.

4.3. Automation Tools and Libraries for Life Cycle Assessment. The major tools that can be used to automate LCA are summarized in the Supporting Information section of the [Supporting Information](#).

5. FUTURE POTENTIAL FOR AUTOMATED LIFE CYCLE ASSESSMENT

There is immense potential for further LCA frameworks that leverage partial automation via AI agents and IAMs. Eventually, it may be possible for both the AI agent and IAM frameworks to conduct full automation, though significant technological advancements are necessary. Despite this promise, there are several shortcomings that need to be addressed, including close adherence to ISO standards, accessibility challenges, computational complexity, and barriers to the adoption of automated LCA.

AI agents also hold great promise for automating pLCA. For instance, the success of AI agents in LCI mapping tasks anticipates potential successes in mapping technologies from an IAM to an LCI database. Moreover, AI agents can aid LCA practitioners by partially automating goal and scope decisions specific to pLCA by recommending policy scenarios or IAMs for use in a pLCA.⁸⁹ These agents could make it easier for LCA practitioners to understand the differences between IAMs and the goals of policy scenarios, offering greater specificity for the justification of scenario choice. Finally, AI agents can leverage the results from an interpretation phase of LCA to automatically suggest new policy scenarios to be modeled in an IAM, increasing the robustness of potential results and enabling LCA practitioners to gain a greater understanding of the sensitivities involved in future production systems. While these opportunities are speculative, potential implementation strategies are given in subsequent subsections.

To avoid needless proliferation of LCA frameworks, it is recommended that not only do users of partially automated frameworks closely adhere to the ISO standards but also that the LCA community work together to establish norms around the use and reporting of automated frameworks, especially for LCAs that involve AI agents due to the potential risks highlighted in [Section 4.1](#). Some AI agents utilize simplifications of the LCA framework for proof-of-concept demonstrations⁸³—this is an important step toward full automation, but the limitations of these simplifying assumptions should be made explicit. For IAM studies, authors highlight the need to describe and justify in detail critical LCA assumptions in adherence with the ISO standards,¹⁵⁶ the relevance of chosen scenarios,¹²⁴ and the relevance of chosen IAMs.¹⁵³ Developing these norms could happen through panels at conferences or similar venues as well as through the peer review process. Furthermore, it may be useful to convene a dedicated task force to provide recommendations of best practices to the LCA community. Standardization of the LCA has been critical for decades and will continue to be important with the development of automated LCA frameworks.

IAMs, AI agents, and LCAs are complex models that are not necessarily easy to understand. As a result, there is a need to increase the accessibility of these tools, to enable their widespread use and ensure that the results from these automated LCA frameworks can be leveraged to improve environmental sustainability. Use of automated LCA frameworks requires significant amounts of knowledge across multiple complex model classes with concomitant understanding of the theory and practice for each of these models. Significant training may be necessary to ensure that LCA practitioners are comfortable using and critiquing studies done using these frameworks. To aid in the adoption of automated frameworks, it would be helpful to document model structures to help translate models between fields¹⁵⁷ and increase accessibility of pLCA tools¹¹ and pLCI databases.¹³² Moreover, AI agent frameworks could also benefit from the development of well-documented repositories,⁹⁵ so that technology can be shared faster. Moreover, making automated tools into a collaborative platform can improve the trustworthiness and reliability of automated LCA.⁸³ To make these automated frameworks more accessible, it is recommended to develop open-source models and for developers to host workshops, provide tutorials, and explore other educational opportunities to teach LCA practitioners how to use these tools. This is especially important because many studies in the supplementary Table S1 do not have automation tools that are publicly available, relying instead on petitions from authors of studies for the data and tools used. Some tools even utilize confidential data that are not available upon request, highlighting the barriers to diffusion of automated tools to LCA practitioners.

In practice, full automation can be quite difficult due to computational challenges. In addition, IAMs and some AI agents based on LLMs require considerable computational resources, most notably RAM, CPUs, GPUs, and time. Most IAMs require significant amounts of memory and processing power when executing a scenario. LLMs require significant GPU usage in the training phase and less GPU usage when in the inference stage or embodied in an AI agent. Only LCA practitioners with access to considerable computation resources will be able to develop these frameworks or at least until applications are created to provide an accessible user interface to less-knowledgeable users. To avoid the purchase of a dedicated high-performance computer, LCA practitioners can take advantage of cloud computing or Web-based interfaces to lower the cost of running these models. Furthermore, developments in reducing the computational requirement of LLMs are common,¹⁵⁸ though this avenue is mostly unexplored for IAMs.¹⁵⁹ Making these frameworks accessible and practical is necessary for them to become widely available and recognized as useful for automation.

Uptake of automated LCA tools, outside of premise, has been minimal. Supposing that automated LCA is conducted at sufficiently high accuracy and rigor to convince the authors and stakeholders of a study, there is still a need to convince other LCA experts, third-party auditors, and the general public about the safety and trustworthiness of these systems. Requiring automated LCA systems to report justifications for decisions is a plausible first step, but convening focus groups and conducting other social research will be necessary to at least identify what information is necessary to trust that an LCA has been conducted properly. Because this research may be outside the existing expertise of LCA practitioners, collaboration with

social scientists may be necessary to better understand the nature of this problem and its potential solutions.

In the Supplementary Discussion section of the [Supporting Information](#), four perspectives on the future of automated frameworks are given, with suggestions for implementation. That section also contains discussion points relevant to the proliferation of scenarios in pLCA.

6. CONCLUSION

LCA automation is feasible using both AI agents and IAMs. Of the tools available for automation, only premise is widespread throughout the pLCA community, in part due to the novelty and perceived untrustworthiness of AI agents. AI agents, although they exist, are not commonly included in LCA workflows. Because of the lack of adoption of existing tools and the speculative nature of potential automated LCA workflows, it is difficult to unilaterally synthesize guidelines and support for the use of automated LCA. As a result, we advocate for a democratic process to establish norms surrounding use in the community through a process of trial and error and may necessitate collaboration with social scientists to understand the opinions of a diverse set of LCA practitioners. To ensure that LCA practitioners are familiar with these tools, efforts should be made to increase their accessibility through training and other educational opportunities. Because these new tools may be viewed with distrust, every opportunity should be taken to justify methodological rigor and adherence to the ISO LCA standards.

These tools are expected to have differing levels of success based on sector. For sectors that can utilize environmentally extended input–output tables or building information modeling, automation using AI agents is easier because the system boundaries, bills of materials, and LCI databases are well-defined and comprehensive. Automating process-based LCA using AI agents is harder as data may not be available for many products or processes. For any study involving novel products or production methods, automation may be less applicable due to a lack of data and precedent in the literature, which will compromise the effectiveness of automation through AI agents. Automating LCA using IAMs could lead to high-quality pLCA for many sectors, so long as they have representation in the IAM. Niche sectors, like pesticides or pharmaceuticals, which are not present in IAMs, cannot be automated using IAMs.

Apart from the four perspectives outlined above, there are more opportunities for future work for automated LCA. For AI agents, more attention could be paid to using multimodal or reinforcement models as the basis of AI agents. For IAMs, increasing the granularity of the coverage is necessary to improve the comprehensiveness of automated pLCA. The inclusion of other environmental impact categories in IAMs will enable the analysis of policy scenarios that are not climate-focused, facilitating a more sustainable future. A study conducting an LCA of various LCA processes will facilitate a better understanding of the environmental trade-offs of using autonomous LCA tools. For any automated LCA, significant work needs to be done to gain the trust of LCA practitioners and validate model performance, in addition to the development of norms surrounding the use of automated tools in an LCA.

Incorporating automation tools into the current LCA workflow is rather difficult. If a pLCA is being conducted, the recommendation is to rely on premise for handling the

pLCA, at least until additional tools are developed to enhance IAMs with LCA. For more specialized tasks, wurst, pathways, superstructures, THEMIS, or ENBIOS can be leveraged. If the goal is to leverage SSP and RCP pathway combinations or custom policy scenarios that are not available on premise, it is likely that a custom tool will need to be developed, though it is recommended to build on the existing software tools mentioned above. Despite the review containing case studies that use AI agents to automate the near totality of all LCA tasks, the software is mostly unavailable. Living Sustainability offers automation support for the whole LCA with simple system boundaries, and Parakeet offers automated impact factor matching for the LCI stage. For all other LCA tasks, custom tools or collaboration with the author of the study to implement a specific automation tool will be necessary. Because there are very few publicly available tools, it may be necessary for LCA practitioners to partner with experts in other fields to develop the tools and make them broadly accessible. When using automated tools, it is important to adhere to methodological rigor, keep humans in the loop (for now), and report justifications for all decisions in the LCA; report the tool used, its inputs and outputs, and the specific tasks that it is automating; make the code used publicly available; and evaluate the tool's performance compared to humans using qualitative analysis and/or benchmarking on LCA task data sets.

Automating LCA can enable scalable and rapid environmentally sustainable decision-making across industries, and there are many pathways to partial and full automation. In contrast to full automation using AI agents, integrating LCA into an IAM requires the development of a single specialized model for pLCA, while an LCA practitioner can choose which AI agents they wish to use. IAMs and AI agents are not mutually exclusive methods for automation—research should proceed while the LCA community identifies standards for the appropriate use of these tools. While full automation of LCA is not yet upon us, the frequency of partially automated frameworks will continue to increase, necessitating further understanding of when, why, and how to use these frameworks, given their advantages and disadvantages.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.5c14493>.

Additional methods for the literature review, quantitative results for the literature review, summary tables for the different AI agent and IAM tools to automate LCA, discussion points on the potential for increase in the number of scenarios when combining LCA and IAM, and discussion of potential designs of automated LCA frameworks (PDF)

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Notes

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